

1. Introduction to Containers: Why Docker?

◆ What Are Containers?

Containers are lightweight, portable, and self-contained software environments that include everything needed to run an application — code, runtime, system tools, libraries, and settings. They ensure that applications run consistently across different environments, whether on a developer's laptop or in production.

◆ Why Do Containers Matter?

- **Consistency:** "Works on my machine" issues are eliminated.
- **Speed:** Containers start quickly and use fewer system resources.
- **Isolation:** Each container runs independently, avoiding conflicts.
- **Scalability:** Easy to replicate and scale applications.

◆ Virtual Machines vs. Containers

Feature	Virtual Machines	Containers
Size	Heavy (GBs)	Lightweight (MBs)
Boot Time	Minutes	Seconds
Resource Usage	High (full OS per VM)	Low (shares OS kernel)
Isolation	Full OS isolation	Process-level isolation
Portability	Less portable	Highly portable

◆ Why Docker in DevOps?

Docker is the most popular containerization platform because:

- It simplifies application packaging and distribution.
- It integrates smoothly with CI/CD pipelines.
- It improves developer productivity.
- It enables easy testing and rollback of app versions.